

MASTER OF PHARMACY (M. PHARM.):

PHARMACOGNOSY

(Two year post graduate degree course)

CREDIT BASED SEMESTER SYSTEM

SCHEME OF TEACHING, EXAMINATION AND SYLLABUS

M. Pharm. I Semester

w.e.f. ACADEMIC SESSION 2013-2014



RKDF University

Bhopal (MP) 462033, INDIA



M. Pharm : Pharmacognosy

Scheme of Teaching and Examination

M. Pharm I Semester Scheme									
Subject Code	Name of Subject	Teaching scheme (hrs/week)		Credits		Examination Scheme			
		T	P	T	P	T		P	
						Ext	Int	Ext	Int
MPMI101[T]	Modern Instrumental Methods	4	-	4	-	70	30	-	-
MPMI101[P]	Modern Instrumental Methods	-	6	-	3	-	-	60	40
MPCG102[T]	Natural Products	4	-	4	-	70	30	-	-
MPCG102[P]	Natural Products	-	6	-	3	-	-	60	40
MPPD103[T]	Product Development and Quality Assurance	4	-	4	-	70	30	-	-
MPPD103[P]	Product Development and Quality Assurance	-	6	-	3	-	-	60	40
MPCG104[T]	Neutraceuticals and Herbal Medicines	4	-	4	-	70	30	-	-
		16	18	16	09	400		300	
Total		34 hrs/week		25		700			

T- Theory, P- Practical, Ext- External, Int- Internal

Theory: 1 Theory hour = 1 Credit

Practical: 2 Practical hour = 1 Credit

Internal assessment (Theory): Best one out of two sessional per semester.

Internal assessment (Practical) Based on day to day performance including attendance, *viva-voce* and laboratory record.



Course	M. Pharm	Semester	First
Branch	Pharmacognosy	Duration	60 Hrs [Theory]
Subject Code	MPMI101[T]	Subject Name	Modern Instrumental Methods

Syllabus

Ultraviolet – Visible spectrometry: Introduction, the nature of electronic excitation, the origin of UV band structure, principle of absorption spectroscopy, chromophore- $\sigma \rightarrow \sigma^*$, $\eta \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$ transitions, basics of instrumentation techniques, pharmaceutical applications. Woodward - Fisher rule for calculation of $\lambda_{\max}$. Introduction to optical rotatory dispersion and circular dichroism. Derivative spectroscopy.

Fourier Transform Infrared Spectroscopy: Introduction, the infrared absorption process, the modes of vibrations, stretching and bending, bond properties and absorption trends, basics of instrumentation techniques, pharmaceutical applications. Interpretation of Infrared spectra.

Nuclear Magnetic Resonance Spectroscopy: High resolution ^1H and ^{13}C NMR spectroscopy, theoretical calculation of chemical shift of various carbon atoms, techniques used for finding out types of carbon attached proton test (APT) distortion less energy polarization transfer (DEPT), homonuclear and heteronuclear correlation spectroscopy. Different 1D and 2D NMR correlation spectrophotometric techniques used as COSy, NOESY, HETCOR, INADEQUATE, HSBC, HMQC, etc. Use of this technique in determination of absolute configuration.

Spectrometry of other important nuclei: Introduction to ^{15}N , ^{19}F , ^{31}P , basic concepts.

Mass spectrometry: Basic principle and theory involved, basics of instrumentation techniques, tandem mass spectrometry and its applications.

HPLC: Instrumentation covering detailed discussion of pumps, injector system, columns and detectors. Analytical method development, validation as per ICH guidelines and troubleshooting. Quantification methods used in HPLC. Ultra pressure liquid chromatography.

HPTLC: Basic instrumentation and its calibration. Analytical method development and its validation as per ICH guidelines. Quantification using HPTLC.

Thermoanalytical techniques: Differential Scanning Calorimetry (DSC), Thermogravimetry (TG), Thermo mechanical analysis (TMA): Principles instrumentation and applications (including interpretation of data) in pharmacy.



Radio analytical techniques used in pharmaceuticals: Isotopic dilution methods, Radioimmunoassay, ELISA etc.

Microscopy: Transmission electron microscopy (TEM), Scanning electron microscopy (SEM), cryomicroscopy, Atomic force microscopy (AFM), confocal microscopy.

Statistics : Collection and classification of experimental data and its statistical treatment, Probability- definition and laws of probability, regression and correlation, method of least squares, correlation coefficient and multiple regression, Test of significance and t-test, statistical quality control, process control, control chart, acceptance sampling plans.

Suggested Readings/Books:

1. Instrumental methods of chemical analysis by chatwal. K, Anand, 5th edition.
2. Organnic spectroscopy by Y.R.Sharma.
3. Text book of pharmaceutical analysis by S.Ravishankar.
4. Spectrometric identification of Organic Compounds, Robert. M. Silverstein et al, 7th Edition, 1981.
5. Principles of Instrumental Analysis by Donglas A. Skoog, James, J. Leary, 4th Edition.
6. Pharmaceutical Analysis – Modern Methods – Part A, Part B, James W. Munson –2001.
7. Practical Pharmaceutical Chemistry, Part two, A. H. Beckett & J. B. Stenlake – 4th Edition.
8. Instrumental Methods of Chemical Analysis – B. K. Sharma - 9th Edition.
9. Chromatography – P. D. Sethi, Dilip Charegaonkar, 2nd Edition.
10. Spectroscopy of Organic Compounds by P. S. Kalsi.
- 11.Organic Spectroscopy by William Kemp
- 12.Pharmaceutical drug analysis by Ashouthosh Kar
- 13.Quantitative analysis of Drugs and Formulations by Sethi



Course	M. Pharm	Semester	First
Branch	Pharmacognosy	Duration	120 Hrs [Practical]
Subject Code	MPMI101[P]	Subject Name	Modern Instrumental Methods

Practicals:

Practical exercises based on the topic mentioned in theory syllabus.



Course	M. Pharm	Semester	First
Branch	Pharmacognosy	Duration	60 Hrs [Theory]
Subject Code	MPCG102[T]	Subject Name	Natural Products

Syllabus

Scope of plant drug cultivation, good agricultural and collection practices, processing of medicinal plant material and problems of cultivation.

Introduction to important properties of soils: General constitution of soils, soil texture and soil structure, characteristics and behaviour of soil water, practical significance of organic and inorganic soil colloids, importance of organic matter of soil, essential elements for plant growth.

Integrated pest management, scope of biological control and use of environment friendly pesticides of plant origin. Brief account of pheromones and juvenile hormones.

Understanding biochemical and physiological aspects of different classes of plant growth regulators. Effect of growth hormones on production of secondary plant metabolites.

Cultivation and management of higher yielding plants especially dioscorea, cinchona, opium, senna, plantago, mentha and lemon grass.

Methods of investigation of biosynthetic pathways; such as tracer technique and auto radiography.

General biosynthetic, studies of alkaloids of pyridine piperidine, tropane, quinoline, isoquinoline, indole and phenthrene types of alkaloids of pharmaceutical significance.

Biosynthesis of steroids cardiac glycosides, flavanoids, coumarins.

Detailed and comparative studies of chemical constituents of drugs are expected. Distribution, detection, extraction, isolation and evaluation of yinca, opium, ergot, rauwolfia, cinchona, digitalis, senna, dioscorea, glyccrihiza, podophyllum, taxus, guggul and artemisia.

Screening of drugs for pharmacological activity - Protocols and screening methods for antidiabetic, antiinflammatory, antihepatotoxic, antifertility and diuretic activity. An overview of current status of plants used as anticancer, antihepatotoxic, antimalarial, antihypertensive and hypolipidemic and adoplogenic agents. Important drugs affecting C.N.S. system.

Herbs and Health foods, Herbal cosmetics, aromatherapy, plants in alternative system of medicine.

Suggested Readings/Books:



1. Trease and Evans “Pharmacognosy”, W.B. Saunders Publication –2002.
2. Varro, E. Tyler, “Pharmacognosy”, 9th Edition, Lea & Febiger – 1998.
- 3 Wallis, T.E. “Text Book of Pharmacognosy”, 5th Edition, CBS Publishers and Distributors – 2002.
4. Cutler, G. Horace “Biologically Active Natural Products Agrochemicals, CRC Press, New York.
5. Sim, Medicinal Plant Alkaloids.
6. Sim, Medicinal Plant Glycosids.
7. Wagner, Economic and Medicinal Plants Vol. I to IV.
8. Wagner, Pharmacognosy, Phytochemistry and Medicinal Plants.
9. Schwartz, Screening Methods in Pharmacognosy.
10. Harborne JB. Phytochemical Methods. Chapman and Hall, London. Latest Edition.
11. Anonymous, WHO Guidelines on Good Agricultural and Collection Practices (GACP) for Medicinal Plants, WHO, Geneva, 2003 or later.
12. Brady NC and Weil RR. The Nature and Properties of Soils. Pearson Education India. Latest Edition.



Course	M. Pharm	Semester	First
Branch	Pharmacognosy	Duration	120 Hrs [Practical]
Subject Code	MPCG102[P]	Subject Name	Natural Products

Practicals:

Practical exercises based on the topic mentioned in theory syllabus.



Course	M. Pharm	Semester	First
Branch	Pharmaceutics	Duration	60 Hrs [Theory]
Subject Code	MPPD103[T]	Subject Name	Product Development and Quality Assurance

Syllabus

Preformulation Studies: Introduction, goals of preformulation, physicochemical properties, criteria for selection of drug and excipients, compatibility tests. Partition coefficient, Solid state pharmaceutics.

Optimization: Definition, need, advantages, Meaning of general terms involved in optimization process. Classification of optimization methods.

Brief description and importance of experimental design with special reference to designs adequate for large number of variables. Introduction of correlation & regression analysis & mathematical model, contour plots. Basic understanding with at least one example of following optimization techniques:-Simplex method, langarengian method, EVOP, Grid search method.

Solubility and Solubilization: Development of theoretical relationships of prognostic relevance, theory and techniques of solubilization of drugs including surfactant systems, co-solvents, solid state manipulations, complexation and chemical modifications, methods of solubility enhancement, factors influencing solubility.

Designing of Pharmaceuticals: Tablet formulation, Special tablets, preparation of components for compression, characterization of granulation, coating of tablets, evaluation of tablets, Equipments and processing problems in tablet.

Liquids: Formulation considerations of oral liquids, suspensions, emulsions, development of various products. Formulation consideration of parenteral, ophthalmic, depot products, large volume and small volume parenterals, environmental control and quality assurance in parenterals.

Disperse systems: Molecular dispersion, coarse dispersions- Physical stability of suspensions and emulsion, role of zeta potential in stability of coarse dispersions, theory of emulsification, micro and multiple emulsions, rheology of suspensions and emulsions. Drug kinetics in coarse disperse systems, drug diffusion in coarse dispersion systems.

Topical and rectal formulations and evaluation.



Introduction to Controlled and Novel Drug Delivery Systems: Sustained release dosage forms, Prodrug concept, nanoparticles, liposomes, resealed erythrocytes, transdermal and other novel drug delivery systems.

Pilot-plant and scale-up techniques.

In vitro and In-Vivo evaluation techniques, product formulation and cGMP

Validation: Concepts in validation, prospective, concurrent, retrospective validation & revalidation, validation of manufacturing, analytical and process validation and its application.

Basic concepts of quality control and quality assurance systems, source and control of quality variation of raw materials: containers, closures, personnel, environmental etc.

In-process quality tests, IPQC problems in Pharmaceutical industries. ICH guidelines.

Sampling Plans, Sampling and Characteristic Curves.

Master formula generation and maintenance, standard operating procedure (SOP) for different dosage forms.

Suggested Readings/Books:

1. Lachmann, L., Lieberman, H.A. & Kanig, J.I.: The Theory and Practice of Industrial pharmacy. Lea and Fibiger, Philadelphia.
2. Banker, G.S. & Rhodes, C.T. : Modern Pharmaceutics, Marcel Dekker Inc. New York and Basel.
3. Turco, S. & King R.E. : Sterile Dosage Forms, Lea and Febiger, Philadelphia.
4. Bean, H.S., Backett, A.H. & Carless, J.E: Advances in Pharmaceutical Sciences, Academic Press, London and Newyork.
5. Jain, N.K.: Controlled and Novel Drug Delivery, CBS, Delhi.
6. Robinson, J.R. & Lee, V.H.L.: Controlled Drug Delivery, Marcel Dekker, New York and Basel.
7. Chien, Y.W.: Novel Drug Delivery Systems, Marcel Dekker, New York and Basel.
8. Jain N. K. Pharmaceutical Product Development, CBS Publisher, Delhi



Course	M. Pharm	Semester	First
Branch	Pharmaceutics	Duration	120 Hrs [Practical]
Subject Code	MPPD103[P]	Subject Name	Product development and quality assurance

Practicals:

Practical exercises based on the topic mentioned in theory syllabus.



Course	M. Pharm	Semester	First
Branch	Pharmacognosy	Duration	60 Hrs [Theory]
Subject Code	MPCG104[T]	Subject Name	Nutraceuticals and Herbal Medicines

Introduction to herbs and ayurveda: Concept of tridosha, role of pancha karma in ayurveda, origin of Charka samhita, comparison of ayurveda with other systems of healing like unani, siddha, homeopathy and yoga.

Herbal product formulations for skin and healthcare, brain tonics and anti-aging products; their manufacturers and suppliers; export market hub of herbs.

Nutraceuticals: General introduction, classification of nutraceuticals, inorganic mineral supplements, vitamin supplements, dietary fibres, antioxidants, health drinks, natural sweeteners, cereals and grains, polyunsaturated fatty acids.

Herbal cosmetics: Formulation and manufacturer of hair dyes, face wash and soaps, skin care creams and lotions, anti-aging creams, shampoos, perfumes, mouthwash, toothpastes.

Herbal medicines for diseases like asthma, Alzheimer's disease, dengue fever, diarrhoea, epilepsy, goitre, gout, hypertension, jaundice, leukemia, obesity, and ulcer of stomach, eczema and diabetes.

Herbal pesticides: General introduction, methods of pest control, manufacturer and exporter of eco guard crop protector - biodegradable pest control products, natural organic pesticides and insecticides and their benefits over chemical pesticides.

Definitions, Nutrients/Nutraceuticals/Dietary Supplements.

Enhancement of General "Well-Being" and Indian Medicinals for Impotence and Diminished Sexual Performance

Remedies for Fatigue, Agents for Enhancement of Physical Performance, Rheumatoid Arthritis and Osteoarthritis.

Herbs used with Respiratory Tract Problems, Diabetes, Antioxidants and Human Health Benefits.

FD&C Act Misbranding

Nutrition Labeling Laws and Regulations Definition of NRC-RDA/U.S. RDA/RDI/DV/DRI

Label Claims, Nutrient Content Claims, Health Claims

Dietary Supplement Health and Education Act 1994: Regulations and Claims

Current Products, Vitamins, Minerals and Herbs



Dietary Fiber and Fiber-like Ingredients: Physiochemical Metabolic Effects, Dairy ingredients as functional foods.

Fats and Oil-Omega-3-6-9 and Fat Substitutes

Nomenclature, Sources, Function and Relationship to Disease

Plant stanols Carotenoids, Antioxidants and Phytochemicals and Their Role in Prevention of Specific Diseases.

GMPs and Shelf life of dietary supplements and functional foods

Soy as a Source of Nutraceutical Components

Bioavailability and safety issues of functional foods

Suggested Readings/Books:

1. Leyel, C. F. 2004. Types of Herbs. Srishti Book Distributors, New Delhi (India).
2. Singh, M. P. and Panda, Himadri 2005. Medicinal herbs with their formulations. Volume 1 & 2. Daya Publishing House, Delhi.
3. Murthy, N. A. and Pandey, D. P. 1998. Ayurvedic Cure for Common Diseases. Orient Paperbacks, Delhi.
4. Rangari., V. D. 2003. Pharmacognosy and Phytochemistry. Vol. II. Career Publications, Nashik.
5. Ojha D. and Kumar, A. 1978. Pancha Karma Therapy in Ayurveda. Chaukhamba Amarabharti Prakashan, Varanasi.
6. Sharma, S. 1983. The System of Ayurveda. Neeraj Publication House, Delhi.
7. Verma, H. K. 1998. A Comprehensive Book of Ayurvedic Medicine for General Practitioners. Kalyani Publishers, New Delhi.
8. Nadkarni, K.M. 1998. India Materia Medica Vol. II. Popular Book Depot, Bombay 7, Dhootapapeshwar Prakashan Ltd. Panvel.
9. Barrett, S. and Herbert, V. 1994. *The Vitamin Pushers*. Prometheus Books, Amherst, NY.
10. Barrett, S. and Jarvis, W. T. 1993. *The Health Robbers*. Prometheus Books, Amherst, NY.
11. Bauernfeind, J. C. and Paul A. LaChance 1991. *Nutrient Additions to Food*. Food & Nutrition Press, Inc. Trumbull, Connecticut.
12. G. Gibson and C. Williams Editors 2000 *Functional foods* Woodhead Publ. Co. London.
13. Goldberg, I. *Functional Foods*. 1994. Chapman and Hall, New York, New York.



14. Labuza, T.P. 2000 Functional Foods and Dietary Supplements: Safety, Good Manufacturing Practice (GMPs) and Shelf Life Testing in *Essentials of Functional Foods* M.K. Schmidl and T.P. Labuza eds . Aspen Press.
15. Lee, Chi-Jen. 1993. *Development and Evaluation of Drugs*. CRC Press, Inc. Boca Raton, FL.
16. Rombeau, J. L. and Michael D. Caldwell. 1990 *Clinical Nutrition: Enteral Nutrition*. 2nd edition W. B. Saunders Company.
17. Schmidl, M.K. and Labuza, T.P. 2000 Medical Foods in *Current Issues in Food Safety* R.Schmidt and Gary Rodrick Editor s J. Wiley and Sons. NY
18. Schmidl, M . K. and Labuza, T.P. 2000 US legislation and functional food claims in *Functional foods* G. Gibson and C. Williams Editors Woodhead Publ. Co. London
19. Schmidl, M . K. and Labuza, T.P. 2000 Nutrition and Health Claims in *Food Labeling* J. Ralph Blanchfield Editor Woodhead Publ. Co. London
20. Schulz, V., Hansel, R., Tyler, V. 1998. *Rational Phytotherapy. A Physicians' Guide to Herbal Medicine*. Springer-Verlag Berlin Heidelberg New York.
21. Shils, ME, Olson, JA, Shike, M. 1994 *Modern Nutrition in Health and Disease*. Eighth edition. Lea and Febiger.
22. Tyler, V. E. 1993. *The Honest Herbal. Pharmaceutical Products Press* New York, NY .



RKDF University

MASTER OF PHARMACY (M. PHARM.):

PHARMACOGNOSY

(Two year post graduate degree course)

CREDIT BASED SEMESTER SYSTEM

SCHEME OF TEACHING, EXAMINATION AND SYLLABUS

M. Pharm. II Semester

w.e.f. ACADEMIC SESSION 2013-2014



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Bhopal (MP) 462033, INDIA



M. Pharm : Pharmacognosy

Scheme of Teaching and Examination

M. Pharm II Semester Scheme									
Subject Code	Name of Subject	Teaching scheme		Credits		Examination Scheme			
		T	P	T	P	T		P	
						Ext	Int	Ext	Int
MPCG201[T]	Plant Drug Standardization	4	-	4	-	70	30	-	-
MPCG202[T]	Plant Biotechnology	4	-	4	-	70	30	-	-
MPCG203[T]	Herbal Drug Technology	4	-	4	-	70	30	-	-
MPCG204[T]	Phytochemical Techniques	4	-	4	-	70	30	-	-
MPCG205[P]	Pharmacognosy Practical-II	-	18	-	09	-	-	60	40
		16	18	16	09	400		100	
Total		34 hrs/week		25		500			

T- Theory, P- Practical, Ext- External, Int- Internal

Theory: 1 Theory hour = 1 Credit

Practical: 2 Practical hour = 1 Credit

Internal assessment (Theory): Best one out of two sessional per semester.

Internal assessment (Practical) Based on day to day performance including attendance, *viva-voce* and laboratory record.



Course	M. Pharm	Semester	Second
Branch	Pharmacognosy	Duration	60 Hrs [Theory]
Subject Code	MPCG201[T]	Subject Name	Plant Drug Standardization

Syllabus

Definitions: Definitions of a range of medicinal plant materials as noted in Ayurvedic Pharmacopoeia of India (API), United States Pharmacopoeia (USP), European Pharmacopoeia (EP) and documents of European Medicines Evaluation Agency (EMA) and World Health Organization (WHO).

Factors Affecting Quality of Plant Drugs: Moisture, temperature, light, oxygen, living organisms, storage of drugs.

Significance of Important Techniques in Establishing Identity, Purity and Quality of Plant Drugs (as described in different Pharmacopoeias, documents of WHO and EMA): Macroscopic, Organoleptic methods including gross morphology, sampling, preliminary examination and foreign matter, Physico-chemical, Solubility, specific gravity, optical rotation, specific rotation, refractive index, melting point, swelling index, foaming index and bitterness value. Moisture content. Ash values, Extractive values including volatile oil, Qualitative chemical tests, Quantitative chemical tests: Acid value, iodine value, saponification value, ester value, unsaponifiable matter and acetyl value, Microscopic, General microscopy, Histochemistry. Quantitative microscopy: Lycopodium spore method, palisade ratio, stomatal number, stomatal index, veinlet number and veinlet termination number, Microbiological (including microbial limits), Biological evaluation.

Assay methods: Markers, choice of active / analytical marker, sources of markers, Marker analysis / fingerprinting by HPLC / HPTLC / GC / fluorimetry / spectroscopy of single plant drugs / extracts and compound formulations. Chemical assays, Radioimmunoassays. Determination of toxic residues. Pesticide residues. Arsenic and heavy metals.

Single plant drugs: Overview of monographs of API, Indian Council of Medical Research (ICMR), WHO, ESCOP, USP, EP and Pharmacopoeia of People's Republic of China.

Compound formulations: Quality aspects of compound formulations of medicinal plants. Monographs of compound formulations of API, control of excipients, stability indicating parameters.



Suggested Readings/Books:

1. Guidelines for the Assessment of Herbal Medicines - WHO Report, Geneva, 1991, Fitoterapia Vol. LXIII, 105-110.
2. Quality Control Methods for Medicinal Plant Material, WHI/ Geneva 1998.
3. Evans WC. Ed. Trease and Evans, Pharmacognosy. Gopson Papers Ltd., Noida, India.
4. Indian Pharmacopoeia. Latest Edition.
5. British Pharmacopoeia. Latest Edition.
6. Indian Herbal Pharmacopoeia. Latest Edition
7. Ayurvedic Pharmacopoeia of India. Latest Edition.
8. Pullock Mukherjee, Quality Control of Herbal Drugs.



Course	M. Pharm	Semester	Second
Branch	Pharmacognosy	Duration	60 Hrs [Theory]
Subject Code	MPCG202[T]	Subject Name	Plant Biotechnology

Syllabus

Plant tissue culture: Historical perspectives, Types and techniques. Organogenesis and embryogenesis, micro propagation of medicinal and aromatic plants. Nutritional requirement of tissue culture, culture media, growth and metabolism of plant tissue culture. Growth parameters of callus and cell culture. Secondary metabolism in tissue cultures, production of pharmaceuticals Role of plants growth of regulators in tissue culture.

Protoplast culture: isolation of protoplast, Haploid protoplast, protoplast fusion and its scope in quality improvement of drug plants. Germ plasm storage cell immobilization, properties and biosynthetic potential of immobilized systems.

Cryopreservation and retention of biosynthetic potential in cell cultures.

Biochemical conversions -Application of plant cell cultures and micro-organisms -Abberant synthesis.

Genetic engineering with special reference to plant cells and micro-organisms Mutation , Hybridization and dyploides chemodems and artificial prod. of mutations.

Culture Aspects- Variability in drug activity . Review of exogenous and endogenous factor in drug production . Soil and plant growth plant nutrition and their role in drug production

Recent studies on Production of mentha, lemongrass, Cinchona, Vinca, ergot, solanaceous drugs and steroidal precursor.

Phytogeograpy and phytogeographical distribution of medicinal plants with special reference to India.

Suggested Readings/Books:

1. Khan, A. Irfan, "Role of Biotechnology in Medicinal and Aromatic Plants, Vol. I to VIII, Ukaaz Publication – 2002.
2. Vyas & Dixit, "Pharmaceutical Biotechnology", CBS Publisher – 1998.
3. Doyle, Alan & Griffith, J, "Cell & Tissue Culture: Laboratory Procedures in Biotechnology. John Willey & Sons.
4. Gamborg, O.L. & Philip G.C, "Plant Cell, Tissue & Organ Culture: Fundamental Methods, Narosa Publication – 1998.



5. Purohit S.S. “Biotechnology: Fundamentals & Application” Agro Bios India – 2001.
6. Scheepler, A. Judith and Gambier M Rossa “Biotechnological Explorations: Applying the Fundamentals” ASM Press, Washington D.C.



Course	M. Pharm	Semester	Second
Branch	Pharmacognosy	Duration	60 Hrs [Theory]
Subject Code	MPCG203[T]	Subject Name	Herbal Drug Technology

Syllabus

Herbal remedies philosophy concepts and bases of herbal medicine, WHO guideline regarding efficacy safety and toxicity of herbal medicines regulatory requirements as per European community, other regulatory authority, Indian Scenario and US. Potential of exploiting Indian herbal medicines,

Factor effecting herb quality – An overview for procurement and storage of drugs. Quality control considerations, Pesticide residue, heavy metal and microbial contamination.

Exogenous and endogenic factors influencing production of crude drugs plant growth regulators and their application in pharmacy. Disease management of medicinal and aromatic plants.

Profiles for commercial cultivation technology and post harvest care of following medicinal plants- ashwagandha, periwinkle, medicinal yams, ergot, guggal, belladonna, senna, rauwolfia, opium poppy, psyllium, steroid bearing solariums ammimajus, ipecac, datura, aloe, henbane, digitalis, saffron.

Quantitative microscopy as applied drugs evaluation and procedures of microtome sectioning procedure, preparations of biological materials for examination by electronic microscope.

Herbal formulations – Quality assurance in new materials, manufacturing of important herbal powders, granules, capsules, tablets, liquids formulations, gel, creams, ointments and other dosage forms. Stability and Biopharmaceutics considerations drug interactions and therapeutic incompatibilities of herbal constituents / formulations.

Dosage forms in Indian system of medicines- Method of preparations of various Ayurvedic dosage forms, Scope of developing modern manufacturing techniques, approaches for stability and quality control, comparative study of Ayurvedic technology for formulation of dosage forms with modern technology of herbal formulations.

Herbal cosmetics- Formulation consideration, study of cosmetic materials of herbs, hair dyes sunscreen, shampoo, lotion, creams etc.

Natural flavour and flavour constituents- Use of natural flavours as flavouring agents. Concept of aromatherapy, important essential oils in aromatherapy.



Neutraceuticals, cosmeceuticals and functional foods- Health benefits of food supplements and functional foods, phytopharmaceuticals as cosmetics, phytosomes.

Suggested Readings/Books:

1. Patykar, K.D. “Herbal Cosmetics & Ancient India: With Treatise on Planta Cosmetic, Indian Book Centre & Publishers.
2. Asharam Vaidhya, “Herbal Indian Perfumes & Cosmetics”, Indian Books Centre & Publishers.
3. Herbal Cosmetics and Ayurvedic Medicines (EOU)
4. Paranjape P., Herbs for Beauty: Revealing Ayurvedic Treasures.
5. Ayurvedic Pharmacopoeia, Vol. I, II & III, Gowthom Medicone Pvt. Ltd., Chennai.
6. Choudhary R.D., “Practical Approach to Industrial Pharmacognosy, Herbal Drug Industry, Eastern Publishers, New Delhi – 1996.
7. Karnik P., Pharmaceutical Standard of Herbal Plants, Syndicate Publishing Co.
8. Sammbamurthy et al “Medicinal Plants in Industry”, Question Bank, CBS Publishers & Distributors.
9. Atal CK and Kapur BM. Eds. Cultivation and Utilization of Medicinal Plants. R.R.L., Jammu, 1982 and supplements there of.
10. Atal CK and Kapur BM. Eds. Cultivation and Utilization of Aromatic Plants. R.R.L., Jammu, 1982 and Supplements.
11. Chadha KL. Ed. Advances in Horticulture: Medicinal and Aromatic Plants. Malhotra Publishing House, New Delhi. Latest Edition.
12. Bilgrami et al, “Phytochemistry & Plant Taxonomy, CBS Publisher & Distributor, New Delhi.



Course	M. Pharm	Semester	Second
Branch	Pharmacognosy	Duration	60 Hrs [Theory]
Subject Code	MPCG204[T]	Subject Name	Phytochemical Techniques

Syllabus

General methods of extraction, isolation and characterization of bioactive constituents

Different extraction methods including advanced extraction techniques like supercritical fluid extraction, microwave-assisted extraction, ultrasound assisted extraction, solid-phase microextraction including headspace technique. Isolation techniques, Fractionation and solvent partitioning.

Chromatography: General principles, classification, normal and reversed phase, bonded phase, separation mechanisms. Chromatographic separations including column chromatography, vacuum liquid chromatography, flash chromatography (medium pressure liquid chromatography), preparative TLC & HPLC, droplet counter-current chromatography (DCCC), ion-exchange chromatography, centrifugal, counter-current chromatography (CCC), centrifugally accelerated chromatography, gas chromatography.

Purification techniques for isolated phytoconstituents.

Introduction to different techniques of characterization of bioactive constituents.

Suggested Readings/Books:

1. Harvey AL. Ed. Advances in Drug Discovery Techniques. John Wiley and Sons, New York. Latest Edition.
2. Hostettmann K, Hostettmann M and Marston A. Preparative Chromatography Techniques: Applications in Natural Product Isolation. Springer-Verlag, Berlin. Latest Edition.
3. Journal of Chromatography- different issues.
4. Cannell RJP. Ed. Natural Products Isolation. Humana Press, Totowa, New Jersey. Latest Edition.
5. Marston A. and Hostettmann K. Modern Separation Methods. Natural Product Reports (1991). Volume 8, 391-413.
6. Reviews on latest chromatographic techniques in various journals.



Course	M. Pharm	Semester	Second
Branch	Pharmacognosy	Duration	120 Hrs [Practical]
Subject Code	MPCG205[P]	Subject Name	Pharmacognosy Practical-II

Practicals:

Practical exercises based on the topics mentioned in theory syllabus of second semester.



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CREDIT BASED SEMESTER SYSTEM

SCHEME OF TEACHING, EXAMINATION AND SYLLABUS

M. Pharm. III Semester

w.e.f. ACADEMIC SESSION 2013-2014



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Bhopal (MP) 462033, INDIA

M. Pharm : Pharmacognosy

Scheme of Teaching and Examination

M. Pharm III Semester								
Subject Code	Name of Subject	Teaching Scheme (hrs/week)		Credits		Examination Scheme		
		T	P	T	P	T	P	
							Ext	Int
MPCGDIS-I	Dissertation - I	-	36	-	18	-	180	120
Total		36 hrs/week		18		300		

T- Theory, P- Practical, Ext- External, Int- Internal

Practical: 2 Practical hour = 1 Credit

Internal assessment (Practical) Based on dissertation work and *viva-voce*.



Course	M. Pharm	Semester	Third
Branch	Pharmacognosy	Duration	36 Hrs/Week
Subject Code	MPCGDIS-I	Subject Name	Dissertation - I

Research work during third semester.

The examination shall be based on dissertation - I submitted at the end of third semester and presentation in open seminar.



RKDF University

MASTER OF PHARMACY (M. PHARM.):

PHARMACOGNOSY

(Two year post graduate degree course)

CREDIT BASED SEMESTER SYSTEM

SCHEME OF TEACHING, EXAMINATION AND SYLLABUS

M. Pharm. IV Semester

w.e.f. ACADEMIC SESSION 2013-2014



RKDF University



RKDF University

Bhopal (MP) 462033, INDIA

M. Pharm : Pharmacognosy

Scheme of Teaching and Examination

M. Pharm IV Semester								
Subject Code	Name of Subject	Teaching Scheme (hrs/week)		Credits		Examination Scheme		
		T	P	T	P	T	P	
							Ext	Int
MPCGDIS-II	Dissertation - II	-	36	-	18	-	180	120
Total		36 hrs/week		18		300		

T- Theory, P- Practical, Ext- External, Int- Internal

Practical: 2 Practical hour = 1 Credit

Internal assessment: Based on dissertation work and *viva-voce*.



Course	M. Pharm	Semester	Fourth
Branch	Pharmacognosy	Duration	36 Hrs/Week
Subject Code	MPCGDIS-II	Subject Name	Dissertation - II

Research work during third and fourth semester.

The examination shall be based on dissertation - II submitted at the end of Fourth semester and presentation in open seminar.